

Intelligent Empowerment of Optical-Computation Synergy: Signal Processing and Resource Management for Multi-Band Optical Networks

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Abstract This paper presents an AI-driven control architecture for optical-computation synergy in AI data center interconnection. Multi-band optical networks provide interconnect support, while AI enhances signal processing and resource management, further strengthening support for computing. We review recent advances and outline future integrated multi-agent control. ©2026 The Author(s)

Introduction

The sustained growth of large-scale training, distributed inference, and interconnection among geo-distributed artificial intelligence data centers (AIDCs) has imposed higher requirements on network bandwidth, latency, reliability, and scalability [1,2]. Optical networks, particularly multi-band optical networks with enhanced capacity scalability, are therefore emerging as a fundamental substrate for AIDC interconnection [3]. At the same time, AIDC workloads are often characterized by high burstiness, high bandwidth demand, and stringent real-time requirements, while the increasing network scale further complicates transmission optimization, resource scheduling, and system management, rendering conventional rule-based approaches insufficient. In this context, optical-computation synergy is emerging as a key paradigm for bridging optics and computation. Specifically, optical networks provide efficient interconnection and elastic transport for computing systems, while computing capabilities, especially AI-driven methods, in turn enhance optical networks through intelligent decision-making and optimization, thereby forming a bidirectional empowerment mechanism.

To meet the high-bandwidth demands among AIDCs, optical networks are extending the usable spectrum beyond the conventional C band to increase system capacity. However, multi-band transmission intensifies the inter-channel stimulated Raman scattering (ISRS) [4], leading to spectral power tilt, generalized signal-to-noise ratio (GSNR) imbalance, and larger performance disparities across bands, which pose more severe challenges to signal processing and quality of transmission (QoT) assurance. Meanwhile, tighter coupling among spectrum, bands, topology, and services makes resource management and global optimization more difficult. Since most existing methods are developed for single-band scenarios, their modelling assumptions and optimization mechanisms are not extensible to multi-band systems. Consequently, new signal processing and resource management techniques

are needed to achieve effective channel equalization, reliable QoT assurance, and efficient resource utilization.

In this paper, we propose an AI-driven intelligent control architecture for AIDC networks. Specifically, we first analyze the key role of optical interconnects in supporting AIDCs. We then review representative advances and our related studies in AI-driven signal processing and resource management for multi-band optical networks. Finally, we discuss future directions toward integrated control with multi-agent cross-layer collaboration.

AI-Driven Control Architecture for AIDC Networks

To enable optical-computation synergy in AIDC networks, we develop an AI-driven intelligent control architecture for multi-band optical networks, as illustrated in Fig. 1. The architecture is organized around two complementary themes: optics for AI and AI for optics. On the one hand, optical interconnect architectures provide the high-bandwidth, reconfigurable, and energy-efficient connectivity required by AIDC workload; on the other hand, AI-driven methods further enhance signal processing and resource management in multi-band optical networks, thereby improving transmission performance, QoT assurance, and resource utilization. More importantly, these optimizations ultimately strengthen the network support provided to upper-layer computing systems, forming a synergistic loop in which

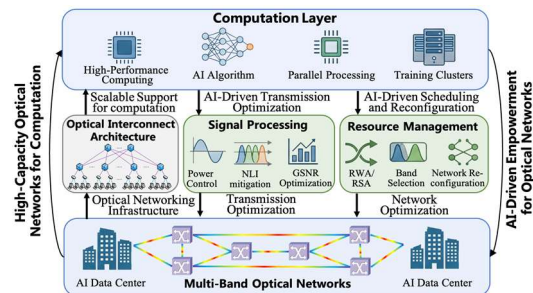


Fig. 1: AI-driven control architecture for optical-computation synergy.

optics supports AI, AI enhances optics, and the resulting improvements further strengthen computation. In the following, we review representative advances and discuss our related studies in these directions.

Optical Interconnect Architecture for AIDCs

In the optics-for-AI direction, recent studies suggest that optical networking is becoming a key infrastructure enabler for large-scale AIDCs. To meet the demands of LLM training for high-bandwidth, low-latency, and energy-efficient interconnects, we propose a reconfigurable optical interconnect architecture based on optical circuit switching (OCS), called “ReFly” [5]. By modeling communication requirements among graphics processing units (GPUs) and combining cycle-decomposition-based network construction with joint scheduling across multiple OCSs, ReFly architecture significantly reduces deployment cost and power consumption while maintaining competitive performance. We developed the distributed evaluation hardware platform shown in Fig. 2(a), which supports 10k-GPU-scale validation of optical-electronic hybrid networks and end-to-end verification with real programmable OCS devices [6,7]. And the experimental results indicate that ReFly architecture achieves 229% higher bus bandwidth than Fat-Tree, as shown in Fig. 2(b). These studies suggest that advanced optical interconnect architectures can provide a network foundation for large-scale AIDCs with higher bandwidth and greater scalability.

AI-Driven Signal Processing

In the AI-for-optics direction, AI-driven signal processing provides more adaptive and effective QoT assurance, thereby offering more stable physical-layer support for AIDC workloads. To address ISRS-induced inter-channel coupling, spectral power imbalance, and performance degradation in multi-band systems, various optimization approaches have been developed. These challenges become more pronounced under intelligent computing scenarios with dynamic traffic and time-varying channel occupancy. Existing works can be broadly categorized into two directions: amplifier-based gain and power control, where Raman amplifiers (RAs) are used for pump configuration and gain shaping; and GSNR-driven transmission optimization, where launch power and amplifier parameters are optimized to improve GSNR and its flatness [8].

In RA-based signal power control, neural networks have been used to learn the inverse mapping from target gain profiles to pump parameters for fast gain control [9]. However, these approaches often overlook the inherent physical constraints of RAs, where the number of controllable parameters is significantly lower than the

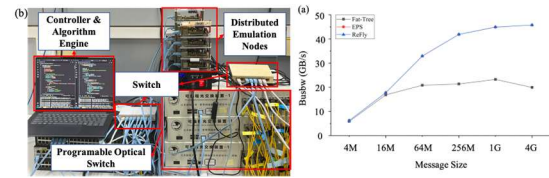


Fig. 2: (a) Distributed evaluation platform. (b) Bus bandwidth comparison of ReFly and benchmarks.

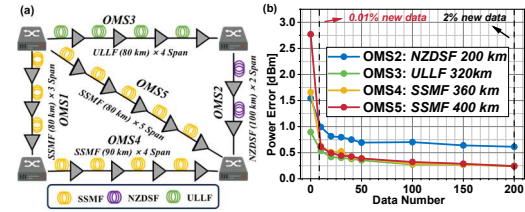


Fig. 3: (a) Experimental simulation setup. (b) Power prediction error versus the number of TL data.

number of wavelength division multiplexing channels. In this case, optimization should adapt to dynamic traffic loads and prioritize service channels, whereas existing methods still rely on static global optimization across all channels, resulting in inefficient use of limited pump resources and reduced control accuracy. To address these limitations, the dynamic adaptive gain optimization (DyAGO) method focuses on service channels with load-aware mechanisms is proposed to improve control accuracy [10]. Furthermore, a fast online optimization approach based on historical models enables rapid calibration in new scenarios using only 1–3 samples, reducing the reliance on large-scale data collection and model retraining during deployment [11].

In GSNR optimization, early methods such as LOGO-based equalization and digital twin (DT)-driven iterative frameworks suffer from high complexity and limited real-time capability. ML-based inverse modelling enables direct optimization, where Transformer models capture inter-channel dependencies and outperform multilayer perceptrons (MLPs) and convolutional neural networks [12]. To improve adaptability across heterogeneous links, an adapter-enhanced Transformer with transfer learning is developed [13]. By freezing the backbone and updating only lightweight adapters, rapid cross-scenario adaptation is achieved with high accuracy. Among different designs, the MLP-based adapter inserted into the encoder stack delivers the best performance.

Fig. 3(a) illustrates the heterogeneous C+L-band optical network used for evaluation, which consists of five optical multiplex sections (OMSs), with OMS1 serving as the source modelling scenario. Fig. 3(b) demonstrates that, with only 2% new data (200 samples), the model achieves efficient transfer, reducing launch power optimization error by ~80% and achieving an average GSNR error of 0.34 dB.

AI-Driven Resource Management

Complementing AI-driven signal processing, AI-driven resource management increasingly focuses on efficient service provisioning and intelligent orchestration for AIDC workloads. Recent studies have therefore moved toward QoT-aware and adaptive resource management. Nakagawa et al. incorporated physical-layer conditions into multi-band resource allocation through wavelength-selective band switching, enabling adaptive C+L-band utilization over hybrid-fiber links [14]. Teng et al. introduced deep reinforcement learning (DRL) into resource allocation for multi-band elastic optical networks, jointly considering QoT constraints and long-term blocking cost [15]. Zhu et al. further employed a hybrid quantum-classical framework to improve the online efficiency of dynamic QoT-aware resource allocation [16]. These studies highlight the importance of QoT-aware provisioning, dynamic scheduling, and intelligent optimization, while also revealing key challenges in topology adaptability and network-wide coordination for AIDC services.

To address these issues, we investigated two complementary directions. First, to improve adaptability to varying topologies and dynamic traffic conditions, we investigated a DRL-based topology-adaptive resource scheduling approach [17], which reformulates service provisioning from conventional path-level selection into a hop-by-hop decision process. By jointly encoding topology, spectrum, and partial-solution states at the node level and using a graph attention network to extract local topology- and spectrum-aware features, the approach improves policy generalization across diverse network structures and traffic scenarios, enabling more efficient online service admission. Second, to address cross-service coupling and network-wide coordination under sustained high-concurrency traffic, we explored a multi-agent Transformer (MAT)-based resource re-optimization scheme [18]. As illustrated in Fig. 4, it uses a Transformer to capture inter-service dependencies induced by ISRS and coordinates multi-service reconfiguration decisions based on global contextual awareness, thereby enabling network-wide resource reorganization. Experimental results show that MAT achieves the highest bandwidth utilization gain across different traffic loads, demonstrating global optimization capability in multi-band environments.

Notably, the synergy between AI-driven signal processing and resource management is not merely additive: improved physical-layer GSNR flatness reduces the uncertainty in QoT estimation, which in turn enables more aggressive and efficient resource allocation at the system level; and finally, enhances the overall support capability of multi-band optical networks for computation.

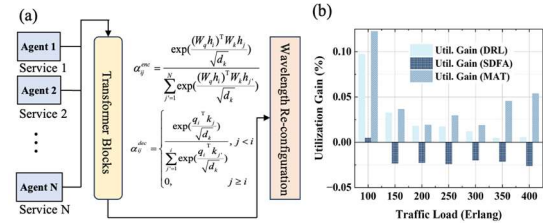


Fig. 4: (a) MAT-based resource re-optimization scheme. (b) Utilization gain versus traffic loads.

Challenges and Future Directions

Current AI-driven methods for optical networks remain decoupled: signal processing focuses on physical-layer optimization, while resource management targets system-level scheduling, lacking a unified feedback loop. However, inter-AIDC traffic is increasingly bursty, dynamic, and tightly coupled, while the state of multi-band optical networks changes continuously with service arrivals, resource occupancy, and physical-layer impairments. In such a setting, the lack of a unified feedback and coordination framework creates a disconnect between local optimization and network-wide objectives, making it difficult to improve gains in overall system efficiency.

A promising direction is to introduce multi-agent architectures into optical networks. The key idea is to unify decision-making processes that are currently scattered across signal processing, resource scheduling, and related functions into a shared framework with cross-layer awareness, joint reasoning, and coordinated execution. Moving in this direction calls for further advances in several aspects. First, more effective cross-layer state representation is needed so that agents can jointly capture physical-layer impairments, resource availability, and traffic dynamics in an informative manner. Second, coordination mechanisms are required to handle the tight coupling among services and optical resources under shared spectrum, band, and topology constraints. Third, more principled credit-assignment strategies are needed to connect local actions with delayed system-level effects. Fourth, scalable real-time control remains essential—it is a cross-cutting challenge that must be addressed in all of the above aspects, ensuring that collaborative decision-making can respond quickly to bursty traffic while preserving global quality. These aspects will be critical to realizing effective multi-agent control in future optical-computation systems.

Conclusions

In this paper, we present an AI-driven control architecture for AIDC networks to support optical-computation synergy, and review recent advances in optical interconnect architecture, signal processing and resource management. In the future, multi-agent collaborative control will show more potential for an adaptive and scalable optical-computation systems.

Acknowledgements

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